**Enqueue and storage**

Fetch packet
64-bit + metadata

Entry-word selection
PC[2] / predicted taken

Ring buffer · 6 × 119 bit
Write: wr / wr+1
Read: rd+2 / rd+3

**Head refill and dual output**

Input bypass

Next0

Head0
119 bit

A

Retained / shifted
heads
Head0 / Head1
Third / fourth word

Candidate validity
Count / PC continuity

Next1

Head1
119 bit

B

**Queue control**

rd / wr / count
Modulo-6

State update
Push 1/2 − pop 0/1/2
Flush priority

Input ready
count ≤ 4

CPU consumption
pop1 ⇒ pop0